

YouTube audience intelligence report - project brief (demo showcase)

What we need built

A pipeline that turns the comment section of any public YouTube video into a short, readable report. You paste a video link into a form, and about a minute later a Google Doc and an email arrive with what viewers liked, what they complained about, the questions nobody answered, and five content ideas taken straight from what the audience asked for. A Slack message carries the headline numbers and a sentiment chart.

It works on our own videos and on anyone else's, which is what makes it interesting in a demo: the prospect names a competitor's latest video and watches their audience get analysed on the spot.

How it works

One workflow, triggered by an n8n form with two fields: the video URL and an email address for the report. Orchestrated by n8n, using the n8n free OpenAI API credits credential:

- **Collect** - a Code node pulls the video id out of the URL. Two HTTP Request nodes call the YouTube Data API (free key): one for the video details (title, channel, views, likes, comment count, publish date), one for the comments with pagination, up to 500 top-level comments and their reply counts, sorted by relevance.
- **Score (gpt-5-nano)** - comments are batched fifty at a time and each gets a sentiment (positive, neutral, negative), a topic tag and a flag for whether it contains an unanswered question. This runs in parallel batches so 500 comments take well under a minute.
- **Analyse (gpt-5.6-luna)** - one call receives the video details, the scored comments and the counts, and writes the report in a fixed structure: one-paragraph verdict, sentiment split with the three most representative comments for each side, recurring themes ranked by frequency, unanswered questions worth replying to, and five content ideas with the comment that inspired each one.
- **Check (gpt-5-mini)** - a reviewer confirms every quoted comment actually appears in the data and that no figure in the report disagrees with the API numbers. On failure the analyse step runs once more with the reviewer's notes; a second failure sends the report anyway, marked as unreviewed.
- **Deliver** - the report is written to a Google Doc in a shared folder, emailed from Gmail as formatted HTML, and summarised in Slack with a QuickChart doughnut of the sentiment split and a link to the Doc. A row is appended to a Google Sheet log with the video, the numbers and the Doc link, so reports build up into a comparable history.

Videos with comments disabled or fewer than twenty comments get a short reply explaining why no report was produced. Quota is not a concern for the demo: the free YouTube API allowance covers roughly a hundred reports a day.

The report (Google Doc)

- Title, channel, publish date, views, likes, comment count, like-to-view ratio
- Verdict - three sentences on how the video landed
- Sentiment - percentages and a bar of positive, neutral and negative, with three quoted comments each
- Themes - five to eight, each with a count and a one-line description
- Unanswered questions - up to ten, verbatim, with a suggested one-line answer
- Content ideas - five, each tied to the comment that prompted it
- Method note - how many comments were read and when

Report log (Google Sheet)

- run_at / requested_by / video_url / video_title / channel
- views / likes / comments_total / comments_analysed
- positive_pct / neutral_pct / negative_pct
- top_theme / doc_link / status - sent / sent_unreviewed / skipped_too_few_comments

Tech stack (all free)

- Orchestration: n8n (self-hosted, already running)
- AI: n8n free OpenAI API credits credential - gpt-5-nano , gpt-5.6-luna , gpt-5-mini
- Data: YouTube Data API v3 (free key from Google Cloud console, 10,000 units a day)
- Output: Google Docs, Gmail and Google Sheets (free, one Google account)
- Chart: QuickChart node (free, built into n8n)
- Notification: Slack (free tier)

What we provide

- Report structure, analysis prompt and reviewer rubric
- YouTube Data API key and a Google account for Docs, Gmail and Sheets
- Slack workspace with a #audience-reports channel
- Ten videos to test against, a mix of our own and competitors', including one with comments disabled
- Access to the existing n8n instance (free OpenAI credits credential already configured)

Deliverables expected

- Working n8n workflows (exported JSON): report generator and error handler
- Ten reports produced from the test list, with the quoted comments spot-checked against YouTube
- Google Sheet log populated and the comments-disabled case handled cleanly
- Slack summary with chart delivered for each run
- Short demo script for the live run, including three competitor videos known to produce strong reports
- Brief handover notes, including how to switch the form trigger to a Slack slash command or a weekly schedule over a channel's newest videos

Timeline

Demo-ready within one week of project start. Collection and scoring on days one and two, report writing and review on day three, delivery and testing on days four and five.

Please quote for the build only. Ongoing maintenance is a separate conversation.

Why this pipeline suits a demo

Quality	What the audience sees
Works on anyone's video	The prospect picks a competitor and gets their audience analysed live
No setup on their side	A public link is the only input
Readable output	A Doc they would actually forward, not a JSON dump
Numbers they can check	Views and comment counts match what YouTube shows
Grounded quotes	Every quoted comment is real and verified by the reviewer step
Builds a history	Each run adds a row, so the sheet becomes a competitor tracker

Source template

Based on the community template "YouTube Video Comment Analysis Agent" (copy in `templates/`), which fetches video details and comments through the YouTube Data API, has one agent write a report, and saves it to Drive and Gmail. This build adds the form trigger, comment pagination to 500, the batched scoring pass, the reviewer check, the QuickChart sentiment chart, the Slack summary and the Sheet log.